Skill Progression Analytics & Performance Benchmarking Report

1. Project Overview

The purpose of this project is to analyze student skill progression and benchmark performance across different colleges, departments, and skill categories. The project was developed using Power BI to transform raw student assessment data into meaningful insights that support data-driven decision-making.

The dashboard helps identify high-performing students, strong skill categories, weak skill areas, and opportunities for performance improvement.

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2. Objectives

The main objectives of this project are:

- Analyze student performance across multiple skill categories.
- Compare assessment scores between departments and colleges.
- Measure completion and failure rates.
- Understand the relationship between learning hours and assessment performance.
- Identify weak skills requiring additional training.
- Generate actionable recommendations for improving student outcomes.

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3. Dataset Description

The dataset contains 1200 student records collected for analysis.

Dataset Fields

- Student ID
- Student Name
- College Name
- Department
- Skill Category
- Skill Name
- Assessment Score
- Learning Hours
- Attempts Count
- Proficiency Level

Dataset Characteristics

- 1200 Student Records
- 5 Skill Categories
- 20+ Skills
- Multiple Colleges
- Multiple Departments

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4. Data Cleaning Process

Before analysis, the dataset was cleaned and validated to ensure accuracy and consistency.

Cleaning Activities

- Checked for missing values.
- Removed duplicate records.
- Verified data types for all columns.
- Standardized categorical values.
- Validated assessment score ranges.
- Verified learning hours and attempts count values.
- Ensured consistency across skill categories and departments.

This process improved data quality and reduced the risk of inaccurate insights.

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5. Data Preparation

After cleaning, the data was organized into a structured format suitable for Power BI analysis.

The preparation process included:

- Formatting column names.
- Verifying relationships between fields.
- Creating proficiency levels.
- Organizing records for KPI calculations.
- Preparing data for visualization and benchmarking.

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6. Power BI Development

The cleaned dataset was imported into Power BI Desktop.

Several DAX measures were created to support dashboard analysis.

Key Performance Indicators

Average Assessment Score

Measures overall student performance.

Total Students

Represents the total number of students included in the analysis.

Completion Rate

Percentage of students achieving passing scores.

Failure Rate

Percentage of students scoring below the passing threshold.

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7. Dashboard Visualizations

The dashboard includes multiple interactive visualizations.

Skill Category Analysis

Analyzes average scores across skill categories and identifies top-performing and low-performing categories.

Department Analysis

Compares student performance across academic departments.

College Benchmark Analysis

Evaluates performance differences between colleges and helps identify benchmark institutions.

Proficiency Distribution

Displays the distribution of Beginner, Intermediate, and Advanced students.

Learning Hours vs Assessment Score

Analyzes the relationship between study effort and academic performance.

Skill Failure Analysis

Highlights skills with lower performance levels and higher failure risks.

8. Key Findings

Overall Performance

- Total Students: 1200
- Average Score: 67.59
- Completion Rate: 77.42%
- Failure Rate: 22.58%

Skill Category Performance

Cyber Security and Cloud Computing demonstrated relatively strong performance compared to other categories.

Programming and Soft Skills showed opportunities for improvement.

Department Performance

Department performance remained relatively balanced, although some departments achieved slightly higher average scores.

College Benchmarking

Certain colleges consistently demonstrated higher average assessment scores, indicating stronger student performance.

Proficiency Levels

A significant portion of students remained at the Beginner level, suggesting a need for additional training and skill development initiatives.

Learning Hours Impact

Students with higher learning hours generally achieved better assessment scores, indicating a positive relationship between effort and performance.

Failure Analysis

Skills such as Linux, Docker, Teamwork, and Security Operations showed lower average scores and require focused improvement efforts.

9. Recommendations

Based on the analysis, the following recommendations are proposed:

Recommendation 1

Increase training hours for low-performing students.

Recommendation 2

Conduct targeted workshops for Programming and Soft Skills.

Recommendation 3

Provide mentoring support for Beginner-level students.

Recommendation 4

Implement regular assessments and feedback sessions.

Recommendation 5

Track student progress through monthly dashboard reviews.

Recommendation 6

Develop specialized learning plans for high-risk skill areas.

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10. Business Impact

The dashboard enables educational institutions to:

- Monitor student performance effectively.
- Identify skill gaps early.
- Improve training strategies.
- Support data-driven academic decisions.
- Increase student success rates.

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11. Conclusion

The Skill Progression Analytics & Performance Benchmarking project successfully transformed raw student assessment data into meaningful business insights using Power BI.

The dashboard provides a comprehensive view of student performance across colleges, departments, and skill categories. Through KPI tracking, benchmarking, and visualization, stakeholders can identify strengths, address weaknesses, and implement strategies to improve overall student outcomes.